FINANCIAL IMPULSE ANALYTICS

Detecting Impulse Behaviour in Young Adults

BEHAVIOURAL ANALYTICS HACKATHON

Presented by Yasir Ahmad

THE CHALLENGE OF IMPULSE BUYING

Problem Statement

Young adults frequently exhibit **overspending tendencies** driven by emotions, peer influence, and seamless "Buy Now, Pay Later" (BNPL) access.

→ Detect impulsive patterns via behavioral markers

→ Identify emotional and promotional triggers

→ Predict high-risk cohorts for intervention

Goal: Quantify abstract psychology into an actionable "Impulse Risk Score".

DATASET & SOURCE OVERVIEW

Parameter	Value / Detail	Context
Total Participants	810 Respondents	E-commerce & BNPL Users
Feature Count	36 Distinct Columns	Psychometric & Behavioural
Primary Source	Juita V, et al. (2024)	Indonesian Consumer Data
Core Framework	SOR Model	Stimulus-Organism-Response

Note: Data specifically contrasts BNPL users against non-users to map triggers to internal emotional states.

BEHAVIOURAL COMPOSITE SCORES

IB_Score

The baseline historical tendency for unplanned impulse buying decisions.

Happy_Score

Current emotional state and euphoria level as a spending catalyst.

Promo_Score

Sensitivity to marketing discounts, flash sales, and e-commerce alerts.

Social_Score

Susceptibility to peer spending

SC_Score

Inherent financial self-control

RISK MODELING ARCHITECTURE

0-100

Normalized Impulse Risk Score

Proprietary Risk Formula

$$Raw_Risk = 1.5 \times (IB_Score) + Happy_Score + Promo_Score + Social_Score + (6 - S$$

→ **Weighting:** Historical impulse given 1.5x significance.

→ **Inversion:** Low self-control mathematically penalizes score.

MODELLING STRATEGY & LOGIC

K-Means Clustering

Unsupervised learning to discover natural behavioral groupings mapped to interpretable personas.

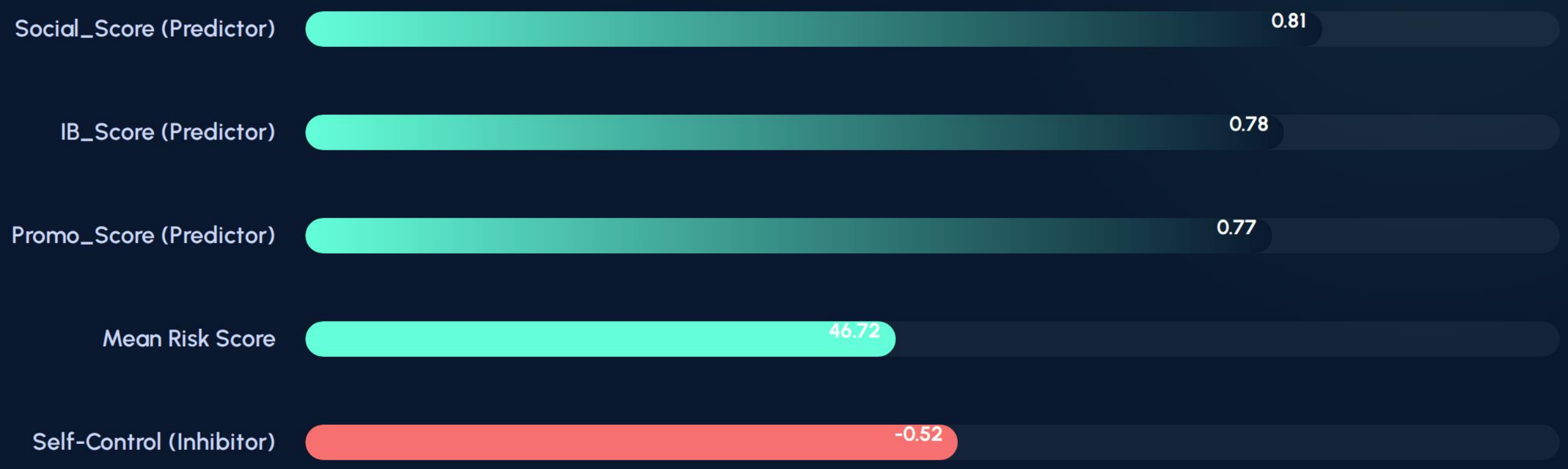
PCA Projection

Principal Component Analysis compressed 5 dimensions into 2, capturing **74.1% variance**.

Statistical Tests

ANOVA and Kruskal-Wallis tests applied to rigorously validate cluster separation significance.

HIGH-LEVEL RISK RESULTS



15.6% of the population (126 people) identified in the Critical High Risk zone (>66 score).

SEGMENTATION & PERSONAS

- The Emotional Spender (434)
- The Deal Chaser (224)
- The Rational Spender (152)

THE THREE SPENDER PERSONAS

The Deal Chaser

High Risk (Mean: 67.19)

Highly promotion-sensitive and socially influenced. Maximum scores on Promo and Social vectors.

The Emotional Spender

Medium Risk (Mean: 44.31)

Largest segment. Spending is mood-driven. Unpredictable habits that fluctuate around the median.

The Rational Spender

Low Risk (Mean: 23.44)

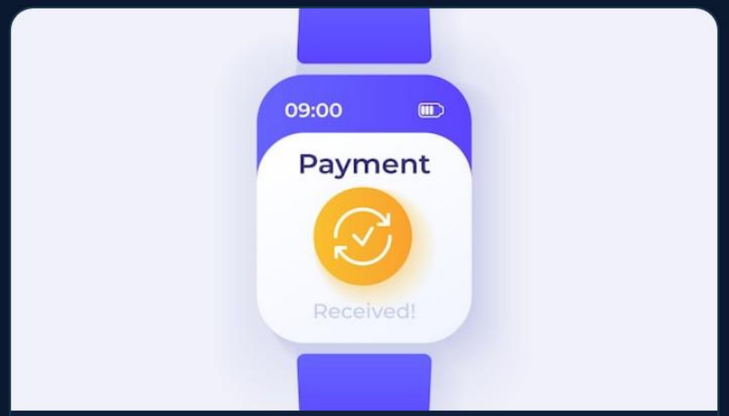
Highly disciplined and budget-conscious. sits strictly below the 25th risk percentile.

STATISTICAL RIGOR & VALIDATION

Factor	F-Statistic	ANOVA p-value	H-Statistic	Significance
IB_Score	292.12	3.62e-96	343.08	**Absolute**
Social_Score	541.54	7.27e-150	452.98	**Absolute**
Promo_Score	525.98	5.92e-147	439.58	**Absolute**

> The infinitesimally small p-values conclusively prove that these behavioral profiles are **distinct realities** and not random variations.

PRACTICAL IMPACT & INTERVENTIONS



Deal Chaser Nudges

Cooling-off periods, budget caps, and impulse blockers for flash sales.

Emotional Spender Tools

Awareness dashboards, spending goal trackers, and aspirational messaging.

Marketing Alignment

Value propositions for Rational Spenders vs Limited-time offers for Deal Chasers.

PRECISION RISK MITIGATION

Questions & Future Scope

Scalable precision-based behavioral risk mitigation. Real-time integration with banking APIs to detect emotional state fluctuations via transaction velocity.

Behavioural Analytics Hackathon | Yasir Ahmad

Data Source: Juita V et al. (2024)

IMAGE SOURCES

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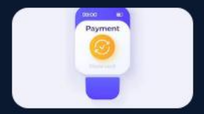

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